

# Venkatesh SM

## Senior DevOps Engineer | Platform Engineering

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### OBJECTIVE

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Senior DevOps & Platform Engineer with 6+ years of experience designing and operating secure, scalable cloud-native platforms on AWS and Kubernetes. Specializes in infrastructure automation, observability, and reliability engineering, driving standardized developer platforms and resilient production systems. Proven track record leading engineers and influencing architecture decisions while improving system reliability, cloud efficiency, and delivery velocity across distributed microservice environments.

### EXPERIENCE

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#### Protium Finance Limited

Bangalore, KA

##### *Senior DevOps Engineer*

*August 2022 – Present*

- Led a **4-engineer platform team** owning **cloud infrastructure, CI/CD, and production reliability**, defining architectural standards adopted across engineering teams.
- Architected and delivered an **RBI-compliant internal GenAI platform** using **OpenWebUI, LiteLLM, and Azure**, enabling secure enterprise LLM adoption while enforcing auditability and data isolation controls.
- Architecting an **AI-driven developer platform** leveraging a **code knowledge graph** to power automated **blast-radius analysis**, vulnerability remediation, and intelligent merge request validation across codebases.
- Implemented **zero-trust access** using **HashiCorp Vault & Boundary**, eliminating static credentials and reducing privileged access risk across production environments.
- Reduced **Kubernetes compute spend by 45%** by migrating production workloads from ASG-based nodes to Karpenter autoscaling on **AWS EKS**, improving resource utilization and scaling efficiency.
- Built an **organization-wide observability platform** (metrics, logs, profiling) using **Grafana, Mimir, Loki, Pyroscope, and OpenTelemetry**, enabling faster RCA and reducing incident diagnosis time.
- Developed **centralized uptime monitoring and OnCall alerting workflows**, improving incident response readiness and production visibility across services.
- Designed and standardized **reusable GitLab CI/CD pipeline templates**, onboarding microservices and Helm-based projects onto a unified delivery workflow, reducing deployment variability and improving developer release velocity.
- Strengthened security posture by remediating **VAPT findings** and integrating centralized **SIEM monitoring with Microsoft Sentinel** for threat detection and audit readiness.
- Integrated **SonarQube quality gates** into CI/CD pipelines, enforcing automated code quality and security checks prior to production deployments.
- Led migration to **Terraform-based Infrastructure as Code** with reusable modules, remote state and CI-gated plan/apply, replacing manual provisioning and standardizing deployments while accelerating environment provisioning.
- Introduced **chaos engineering practices** using **ChaosToolkit** to automate disaster recovery drills and validate system resilience under failure scenarios.
- Operated centralized application logging on the **ELK stack – Fluent Bit** shipping from **Kubernetes** workloads into **OpenSearch**, with **Kibana** dashboards and index lifecycle/retention policies, enabling self-serve log search for engineering teams.

#### Tata Consultancy Services

Chennai, TN

##### *System Engineer - DevOps*

*August 2021 – July 2022*

- Automated infrastructure configuration using **Puppet** integrated with **CI/CD pipelines**, improving consistency of application and environment deployments.
- Managed Linux VM infrastructure lifecycle (provisioning, patching, decommissioning) while introducing standardized deployment workflows.

##### *Assistant System Engineer*

*August 2020 – July 2021*

- Developed and maintained **Puppet manifests** and Hiera configurations to support scalable environment provisioning.
- Supported **MySQL** migration and upgrade initiatives in collaboration with database engineering teams.

## EDUCATION

SRM Easwari Engineering College | Chennai, TN

Bachelor of Engineering, Computer Science

2016 – 2020

GPA: 7.1/10

## SKILLS & INTERESTS

**Platform & Reliability Engineering:** Platform Engineering, Site Reliability Engineering (SRE), Observability, Incident Response, Chaos Engineering

**Cloud Platforms:** AWS (EKS, EC2, S3, IAM, VPC, RDS, ALB, CloudWatch, Route 53, ECR), Azure

**Containers & Orchestration:** Kubernetes, EKS, Docker, Helm, Karpenter

**Infrastructure as Code:** Terraform, AWS CDK

**AI & LLM Platform:** LLM gateway (LiteLLM), OpenWebUI, RAG, model governance & auditability, AIDLC

**CI/CD & Developer Platforms:** GitLab CI, GitHub Actions, Jenkins, Backstage, SDLC

**Observability Stack:** Grafana, Prometheus, Mimir, Loki, Pyroscope, OpenTelemetry, Alloy, Grafana OnCall

**Security & Compliance:** Zero Trust, HashiCorp Vault, Boundary, Microsoft Sentinel, WAF, SonarQube, VAPT Remediation

**Databases & Data Stores:** PostgreSQL, Redis, MySQL

**Programming & Automation:** Bash, Python, Java

**Operating Systems & Networking:** Linux (Ubuntu, RHEL/Amazon Linux), systemd, iptables, DNS, TLS, Nginx

**Interests:** AI-assisted DevOps, System Resilience, Infrastructure Automation

## PROJECTS

### CloudLab - Personal Cloud Infrastructure Platform

- Designed and operated a personal AWS-based cloud environment hosting a finance analytics stack with infrastructure fully managed through Infrastructure as Code.
- Implemented on-demand infrastructure provisioning using **AWS Chatbot**, reducing idle resource costs through automated runtime control.
- Built automated backup and recovery workflows with **S3** and external storage to ensure data durability and disaster recovery readiness.
- **Stack:** Terraform, AWS Chatbot, EC2, ECS, S3

### Configuration Management using Ansible [🔗 GitHub](#)

- Developed reusable **Ansible** automation to provision Linux workstations and servers from a clean state, enabling consistent and reproducible environments.
- **Stack:** Ansible, Shell, Jinja

### Version-Controlled Resume Pipeline [🔗 GitHub](#)

- Built a CI/CD-driven documentation pipeline where resume updates automatically generate and deploy production-ready PDFs.
- Implemented automated publishing using **GitHub Actions** and GitHub Pages.
- **Stack:** LaTeX, GitHub Actions, YAML



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